

THEME: COLLECTIONS + HASHING + ITERATION SAFETY

Objective: Test understanding and analysis of Lists, Sets, Maps, and iteration behavior (Sessions 19–23).

TASK 1: Course Enrollment Using List & Set

Time: 45 mins

Requirements:

- Maintain list of courses using `ArrayList`
- Maintain enrolled students using `HashSet`
- Prevent duplicate enrollments
- Print total enrolled students

Concepts Required:

`ArrayList`, `HashSet`, collection design choice.

Outcome (BTL-3): Apply correct collection selection

Hidden Check: Can student justify List vs Set?

TASK 2: Student Lookup Using HashMap

Time: 40 mins

Requirements:

- Use `HashMap<Integer, Student>`
- Store students by ID
- Retrieve student by ID
- Handle missing ID safely

Concepts Required:

Map usage, key-value modeling.

Outcome (BTL-4): Analyze fast lookup strategy

TASK 3: Iterator & Fail-Fast Experiment

Time: 35 mins

Requirements:

- Create list of students
- Try removing inside for-each loop (observe error)
- Fix using `Iterator.remove()`
- Explain fail-fast behavior

Concepts Required:

Iterator, `ConcurrentModificationException`.

Outcome (BTL-4): Analyze iteration safety

Industrial Insight: Prevent runtime crashes